BEYOND PASS OR FAIL

What the Bechdel Test Can Teach Us About Bayesian Inference

WHO IS THIS WEBSITE FOR?

This website is intended for students who have completed STAT 360 or a similar probability course and are familiar with frequentist methods. You should be familiar with probability distributions, random variables, expected values, and basic statistical notation.

You do not need previous experience with Bayesian statistics. Concepts such as priors, likelihoods, posterior distributions, conjugacy, credible intervals, and posterior prediction will be introduced from the beginning.

We will briefly review frequentist ideas such as maximum likelihood estimation, confidence intervals, and hypothesis testing. The goal is not to reteach all of frequentist inference, but to use familiar methods as a bridge to Bayesian thinking.

THE QUESTION

Think about the last few movies you watched. How many featured at least two women whose names you knew? Did those women ever speak to each other? And if they did, did they talk about anything other than a man?

These questions form the Bechdel test. A movie passes if:

1. It contains at least two named women.
2. The women speak to each other.
3. Their conversation is about something other than a man.

(Try the Bechdel test on a few movies you've watched recently! One of the latest 2026 hit releases, *The Odyssey*, despite having a star-studded female cast, fails the Bechdel test, while *The Wuthering Heights* with its central theme of an intense whirlwind-romance–to say the least–somehow passes the test.)

The Bechdel test is not a complete measure of whether a movie is good, feminist, or free from sexism. A movie can pass while still portraying women poorly, and a movie can fail despite having complex female characters. Instead, the test captures one specific feature of representation: whether women appear as named characters who interact about something beyond a male character.

This simple test leads to a simple curiosity:

How many movies overall pass the Bechdel test?

We cannot watch and classify every movie ever produced so instead, we observe a sample and use it to learn about an unknown population proportion (i.e., the proportion of movies that pass the Bechdel test).

THE MOVIE SAMPLE

To investigate the proportion of movies that pass the Bechdel test, we will examine a sample of 20 movies. Of these movies:

- 9 passed the Bechdel test
- 11 failed the Bechdel test.

The observed pass rate is:
9/20 = 0.45

In other words, 45% of the movies in this particular sample passed. Our goal is to determine what this result can tell us about π, the unknown proportion of comparable movies that pass the test.

SECTION 1: FROM TWENTY MOVIES TO AN UNKNOWN PROPORTION

Let π represent the unknown proportion of movies in the relevant population that pass the Bechdel test.

π = P(a randomly selected movie passes the Bechdel test) *Note: will change pi to p*

If π = 0.30, then approximately 30% of movies in the population pass.

If π = 0.50, then approximately half pass.

If π = 0.80, then approximately 80% pass.

We do not know the true value of π. What we observe is the sample proportion:

$\hat{\pi} = y/n$

Here:

$\hat{\pi} = 9/20 = 0.45$

The sample tells us that 45% of these 20 movies passed. However, $\hat{\pi}$ and π are not the same thing. π is the unknown population proportion we want to estimate and $\hat{\pi}$ is a statistic calculated from the particular sample we observed.

INTERACTIVE PLOT (?)

Choose a hypothetical value of π and generate repeated samples of 20 movies. The graph will display the sample pass rate from each simulation. Even when π remains fixed, the observed sample proportion will change from sample to sample.

SECTION 2: CONSTRUCTING A PROBABILITY MODEL

For each movie, define:

- $X_i = 1$ if movie i passes the Bechdel test
- $X_i = 0$ if movie i fails the Bechdel test

Because each observation has two possible outcomes, we begin with a Bernoulli model:

$X_i \mid \pi \sim \text{Bernoulli}(\pi)$

This notation means that, conditional on a particular value of π:

$P(X_i = 1 \mid \pi) = \pi$

$P(X_i = 0 \mid \pi) = 1 - \pi$

Now let Y be the total number of passing movies:

$Y = X_1 + X_2 + ... + X$□

If the observations satisfy the Binomial assumptions, then:

$Y \mid \pi \sim \text{Binomial}(n, \pi)$

For our sample:

$Y \mid \pi \sim \text{Binomial}(20, \pi)$

and we observed:

$Y = 9$

This model depends on several assumptions:

1. Each movie is classified as either passing or failing.
2. The sampled movies are independent.
3. Each movie has the same probability π of passing.
4. The sample reasonably represents the population about which we want to make an inference.
5. The Bechdel classifications are accurate and consistently applied.

INTERACTIVE PLOT (?)

Move a slider for π and watch the Binomial distribution for Y change. The plot will show the probabilities of observing 0, 1, 2, ..., 20 passing movies under the selected value of π.

SECTION 3: TWO WAYS TO DESCRIBE UNCERTAINTY

Frequentist and Bayesian inference begin with the same data and can use the same Binomial model. Their main difference is how they represent uncertainty about π.

THE FREQUENTIST VIEW

In frequentist inference, π is fixed but unknown. It does not receive a probability distribution.

The sample is random. If we repeatedly selected new samples, the number of passing movies would change. Frequentist methods use this repeated-sampling behavior to evaluate estimators, construct confidence intervals, and perform hypothesis tests.

A frequentist might ask:

- Which value of π makes the observed data most likely?
- Which values of π are reasonably compatible with the sample?
- Would 9 or fewer passes be unusual if half of all movies passed?

THE BAYESIAN VIEW

In Bayesian inference, uncertainty about π is represented using a probability distribution.

The distribution does not mean that the population proportion physically changes at random. Instead, it describes our uncertainty about its fixed but unknown value.

A Bayesian might ask:

- Which values of π seemed plausible before observing these 20 movies?
- How should the 9 passes and 11 failures change that uncertainty?
- What is the probability that fewer than half of movies pass?

The Bayesian analysis has three main ingredients:

Prior distribution × Likelihood → Posterior distribution

- The prior represents uncertainty before the current data.
- The likelihood represents the evidence supplied by the current data.
- The posterior represents updated uncertainty after combining the two.

SECTION 4: THE LIKELIHOOD– WHAT DOES THE SAMPLE SUPPORT?

Under the Binomial model, the probability of observing y passes among n movies is:

$P(Y = y \mid \pi) = C(n,y)\pi^y(1 - \pi)^{(n - y)}$

Here, C(n,y) represents the number of ways that y passing movies can occur among n movies.

Once the data have been observed, y and n are fixed. We then examine the expression as a function of the unknown π. This creates the likelihood:

$L(\pi \mid y) = C(n,y)\pi^y(1 - \pi)^{(n - y)}$

For 9 passes among 20 movies:

$L(\pi \mid y) = C(20,9)\pi^9(1 - \pi)^{11}$

The value C(20,9) does not change when π changes. It affects the height of the likelihood curve, but not its shape or the location of its maximum. Therefore, we can write:

$L(\pi \mid y) \propto \pi^9(1 - \pi)^{11}$

The likelihood asks:

Which possible values of π make the observed result of 9 passes and 11 failures relatively plausible?

Values close to 0.45 receive the greatest likelihood support. Values close to 0 or 1 receive very little support because they make a mixed result of 9 passes and 11 failures unlikely.

However, the likelihood is not automatically a probability distribution for π. It allows us to compare parameter values, but it does not yet allow us to say that there is a certain probability that π lies in an interval.

INTERACTIVE PLOT

Move a proposed value of π across the likelihood curve. The graph will report its relative likelihood and compare it with the value receiving the greatest support.

SECTION 5: A BRIEF FREQUENTIST ANALYSIS

The maximum likelihood estimate is the value of π that maximizes the likelihood:

$\hat{\pi}_{MLE} = \arg\max L(\pi \mid y)$

For a Binomial model:

$\hat{\pi}_{MLE} = y/n$

Therefore:

$\hat{\pi}_{MLE} = 9/20 = 0.45$

The MLE gives us a single best estimate of π, but it does not show how precise that estimate is. To represent the uncertainty caused by sampling only 20 movies, we can construct a 95% Wilson confidence interval:

(0.258, 0.658)

This interval is fairly wide because the sample is small. Based on the observed data, pass rates between approximately 25.8% and 65.8% are reasonably compatible with the sample.

Formally, the 95% confidence level describes the long-run reliability of the Wilson interval procedure. If we repeatedly sampled 20 movies in the same way and constructed an interval from each sample, approximately 95% of those intervals would contain the fixed true value of π.

It does not formally mean that there is a 95% probability that π lies between 0.258 and 0.658. In the frequentist framework, π is fixed, while the interval changes from sample to sample.

Note: We will learn that this differs from a Bayesian 95% credible interval, which allows us to say that, given the prior, model, and observed data, there is a 95% posterior probability that π lies within the reported interval.

We can also test whether the pass rate is below 50%:

H_0: π = 0.50

H_a: π < 0.50

If the null hypothesis were true, then:

Y ~ Binomial(20, 0.50)

The one-sided p-value is:

P(Y ≤ 9 | π = 0.50)

= Σ from k = 0 to 9 of C(20,k)(0.50)^k(0.50)^(20−k)

≈ 0.412

This means that if the true pass rate were 50%, there would be approximately a 41.2% probability of observing 9 or fewer passes in a sample of 20.

That outcome is not particularly unusual. Therefore, the sample does not provide strong frequentist evidence that fewer than half of comparable movies pass.

The p-value is not:

$P(H_0 \text{ is true} \mid \text{data})$

It does not mean that there is a 41.2% probability that π equals 0.50. It measures the probability of the observed result, or something more extreme, assuming the null hypothesis is true.

INTERACTIVE PLOT (?)

The Binomial distribution under $\pi = 0.50$ will be displayed. Bars corresponding to $Y \leq 9$ will be shaded to show which outcomes are included in the p-value.

SECTION 6: BEGINNING THE BAYESIAN ANALYSIS

The frequentist analysis told us that 9 or fewer passes would not be unusual if π were 0.50. However, it does not directly answer the question we may be most interested in:

- What is the probability that fewer than half of comparable movies pass?

To make a probability statement about π itself, we need to represent our uncertainty about π with a probability distribution. This brings us to the Bayesian approach. Unlike the frequentist analysis, however, Bayesian inference requires us to consider what values of π seemed plausible before we observed the sample.

Suppose three moviegoers are asked the following question before seeing the sample: What proportion of movies do you believe pass the Bechdel test?

- The first expects the pass rate to be relatively low.
- The second is genuinely uncertain and does not favor low, middle, or high values.
- The third believes that the Bechdel test sets a low bar and expects nearly all movies to pass.

These people do not begin with the same beliefs. A Bayesian analysis does not require us to hide this disagreement. Instead, each person can represent their uncertainty with a prior distribution.

A prior is not simply one predicted value. It is a distribution over all possible values of π.

For example, someone might think that π is probably near 0.30 but admit that values such as 0.20 or 0.50 are also possible. Another person might place equal prior plausibility on every value between 0 and 1.

The prior makes these starting assumptions visible.

SECTION 7: WHY CHOOSE A BETA PRIOR FOR THE MOVIES?

Because π is a proportion, it must lie between 0 and 1. We therefore need a prior distribution with the same possible values.

The Beta distribution is a convenient choice:

$\pi \sim \text{Beta}(\alpha, \beta)$

Its density is:

$p(\pi) = [1/B(\alpha,\beta)]\pi^{(\alpha-1)}(1-\pi)^{(\beta-1)}$, for $0 < \pi < 1$

The term $B(\alpha,\beta)$ is a normalizing constant. It ensures that the total area under the prior density equals one.

The Beta distribution is useful for two reasons.

First, it is flexible. Depending on α and β, it can favor:

- low values of π,
- high values of π,
- values near 0.50,

or every value equally.

Second, the Beta distribution is conjugate to the Binomial likelihood. Conjugacy means that a Beta prior combined with a Binomial likelihood produces a Beta posterior. The family of the distribution stays the same, although its parameters change.

This makes the update mathematically simple and visually intuitive.

Source: Bayes Rules!, Chapter 3
https://www.bayesrulesbook.com/chapter-3.html

INTERACTIVE PLOT (?)

A gallery will display several Beta distributions. Readers will match each curve to a description such as “uncertain,” “expects a low pass rate,” or “strongly expects a high pass rate.”

SECTION 8: WHAT DO ALPHA AND BETA CONTROL? *Note: do I need this section?*

The parameters α and β determine the location and concentration of a Beta distribution.

Both parameters must be positive:

$\alpha > 0$ and $\beta > 0$

The mean of a Beta distribution is:

$E(\pi) = \alpha/(\alpha + \beta)$

This tells us where the prior is centered.

- If $\alpha = \beta$, the prior mean is 0.50.
- If $\alpha > \beta$, the prior favors higher values of π.
- If $\alpha < \beta$, the prior favors lower values of π.

The total:

$\alpha + \beta$

helps determine the prior's concentration or strength.

A small total generally produces a more widely spread prior. A large total generally produces a more concentrated prior and gives the prior greater influence in the posterior update.

Consider two examples:

$\pi \sim \text{Beta}(2,2)$

$\pi \sim \text{Beta}(20,20)$

Both have a prior mean of 0.50:

$2/(2+2) = 0.50$

$20/(20+20) = 0.50$

However, Beta(2,2) is widely spread, while Beta(20,20) is tightly concentrated near 0.50. The two priors agree about their center but disagree about their certainty.

In a Beta-Binomial analysis, α and β loosely behave like prior pass and failure information. However, they should not automatically be described as literal historical movie counts. In this example, they are parameters chosen to represent different hypothetical viewpoints.

INTERACTIVE PLOT

Readers will adjust α and β with sliders. The graph will update the prior curve, mean, and concentration. A second control will hold the mean fixed while changing $\alpha + \beta$, making it possible to see the difference between the prior's location and strength.

SECTION 9: THREE DIFFERENT STARTING BELIEFS

Consider three priors.

THE LOWER-PASS VIEWPOINT

The first moviegoer believes that most movies probably fail:

$\pi \sim \text{Beta}(5,11)$

The prior mean is:

$E(\pi) = 5/(5+11) = 0.3125$

This prior places most of its density below 0.50, but it does not completely rule out higher pass rates.

THE NEUTRAL VIEWPOINT

The second moviegoer has no strong initial belief about whether the pass rate is low or high:

$\pi \sim \text{Beta}(1,1)$

The prior mean is:

$E(\pi) = 1/(1+1) = 0.50$

Beta(1,1) is equivalent to a Uniform(0,1) distribution. Its density is flat, so it gives equal prior density to every value between 0 and 1.

This does not mean that π has no prior. It means that the prior deliberately avoids favoring one region of the interval.

THE HIGHER-PASS VIEWPOINT

The third moviegoer strongly expects almost all movies to pass:

$\pi \sim \text{Beta}(14,1)$

The prior mean is:

$E(\pi) = 14/(14+1) \approx 0.933$

This prior places substantial density near 1 and little density on low values.

These priors are not equally uncertain. Beta(1,1) is very diffuse, whereas Beta(14,1) expresses a strong and concentrated belief. Therefore, differences in their posteriors will reflect both the locations and strengths of their priors.

INTERACTIVE PLOT (?)

The three prior curves will appear without labels. Readers will predict which curve belongs to each viewpoint and then reveal the answer.

SECTION 10: UPDATING WITH BAYES' RULE

Now we reach the central step of Bayesian inference: combining our prior beliefs with the movie data to produce an updated distribution for π.

The general form of Bayes' rule is:

$p(\pi \mid y) = [p(y \mid \pi)p(\pi)] / p(y)$

This can be summarized as:

Posterior $\propto$ Likelihood $\times$ Prior

Here:

- The prior describes our uncertainty about π before observing the sample.
- The likelihood describes how compatible the observed data are with different values of π.
- The posterior describes our updated uncertainty about π after observing the sample.

The likelihood for the 9 passing and 11 failing movies is:

$L(\pi \mid y) \propto \pi^9(1-\pi)^{11}$

A Beta prior has the following kernel:

$p(\pi) \propto \pi^{(\alpha-1)}(1-\pi)^{(\beta-1)}$

When we multiply the likelihood and prior, we obtain:

$p(\pi \mid y) \propto \pi^9(1-\pi)^{11} \times \pi^{(\alpha-1)}(1-\pi)^{(\beta-1)}$

Combining terms with the same base gives:

$p(\pi \mid y) \propto \pi^{(\alpha+9-1)}(1-\pi)^{(\beta+11-1)}$

This has the same form as another Beta distribution. Therefore:

$\pi \mid y \sim \text{Beta}(\alpha+9, \beta+11)$

More generally, if y movies pass out of n movies:

$\pi \mid y \sim \text{Beta}(\alpha+y, \beta+n-y)$

The update has a memorable structure:

- Posterior α = Prior α + observed passes
- Posterior β = Prior β + observed failures

This calculation shows exactly how Bayesian learning occurs. Each observed pass adds to the first parameter, while each observed failure adds to the second. The resulting posterior combines our starting assumptions with the evidence from the sample.

What is conjugacy?

In general, a prior distribution is called **conjugate** to a likelihood when the resulting posterior belongs to the same family of distributions as the prior. The parameter values change after observing the data, but the type of distribution does not.

In this setting, the Beta distribution is conjugate to the Binomial likelihood. We begin with a Beta prior for π, use a Binomial model for the number of passing movies, and obtain another Beta distribution as the posterior:

Beta prior + Binomial likelihood → Beta posterior

For example, if we begin with the uniform prior:

$\pi \sim \text{Beta}(1,1)$

then observing 9 passes and 11 failures produces:

$\pi \mid y \sim \text{Beta}(1+9, 1+11)$

$\pi \mid y \sim \text{Beta}(10,12)$

Conjugacy is convenient because it allows us to update the prior by changing only its parameters. Because the posterior remains a Beta distribution, we can easily estimate π, construct credible intervals, and calculate probabilities such as:

$P(\pi < 0.50 \mid \text{data})$

Conjugacy is a computational convenience, not evidence that a particular prior is appropriate. We must still explain why the prior is reasonable and examine whether different prior choices meaningfully change our conclusions.

SECTION 11: THREE PRIORS MEET THE SAME DATA

All three moviegoers observe the same evidence:

9 passes and 11 failures.

For the lower-pass viewpoint (i.e., the moviegoer believes most movies will fail the test):

Prior: Beta(5,11) – 5 successes, 11 failures

Posterior: Beta(5+9, 11+11)

Posterior: Beta(14,22)

For the neutral viewpoint (i.e., the moviegoer believes there is a 50/50 chance of success):

Prior: Beta(8,8) – 8 successes, 8 failures

Posterior: Beta(1+9, 1+11)

Posterior: Beta(10,12)

For the higher-pass viewpoint (i.e., the moviegoer believes most movies will pass the test):

Prior: Beta(11, 5) – 11 successes, 5 failures

Posterior: Beta(14+9, 1+11)

Posterior: Beta(23,12)

The data pulls all three people toward the observed pass rate of 0.45. However, the sample is not large enough to erase their initial disagreement.

The lower-pass moviegoer sees the data as reasonably consistent with their original expectations.

The neutral moviegoer allows the sample to dominate because the flat prior contributes little specific information.

The higher-pass moviegoer moves considerably downward from a prior mean of 0.933, but the strong optimistic prior prevents the posterior from moving all the way to 0.45.

INTERACTIVE PLOT (?)

For each viewpoint, the prior, likelihood, and posterior will be displayed together. Readers will choose a viewpoint and watch the posterior form between the prior and the likelihood.

SECTION 12: THE POSTERIOR MEAN AND SHRINKAGE

After finding the entire posterior distribution, we may want a single value that summarizes our updated estimate of π. One common Bayesian estimate is the **posterior mean**.

The posterior mean is the average of all possible values of π under the posterior distribution, weighted by how plausible each value is after observing the data. It can be interpreted as our updated point estimate of the proportion of movies that pass the Bechdel test.

For a Beta(a,b) posterior, the posterior mean is:

$E(\pi \mid y) = a/(a+b)$

Using the lower-pass prior, we obtained:

$\pi \mid y \sim \text{Beta}(14,22)$

Therefore:

$E(\pi \mid y) = 14/(14+22)$

$E(\pi \mid y) = 14/36 \approx 0.389$

Based on this prior and the observed sample, our posterior estimate is that approximately 38.9% of comparable movies pass the Bechdel test.

The sample proportion was:

$y/n = 9/20 = 0.45$

However, the posterior mean is approximately 0.389. Why is the posterior estimate lower than the sample proportion?

The Beta(5,11) prior had a mean of:

$5/(5+11) = 0.3125$

The posterior mean falls between the prior mean of 0.3125 and the sample proportion of 0.45. It has moved away from the sample proportion and toward the prior mean. This movement is called **shrinkage**.

In Bayesian statistics, shrinkage occurs when the posterior estimate is pulled toward the prior estimate. The amount of shrinkage depends on the relative amount of information represented by the prior and the data.

The posterior mean can be written as:

$E(\pi \mid y) = [(\alpha+\beta)/(\alpha+\beta+n)] \times [\alpha/(\alpha+\beta)] + [n/(\alpha+\beta+n)] \times [y/n]$

This shows that the posterior mean is a weighted average of:

Prior mean = $\alpha/(\alpha+\beta)$

Sample proportion = y/n

For the Beta(5,11) prior:

$\alpha+\beta = 5+11 = 16$

The value $\alpha+\beta$ is called the **prior concentration**. In this conjugate model, it controls how much influence the prior has relative to the sample.

The two weights are:

Prior weight = 16/(16+20) = 16/36 ≈ 0.444

Data weight = 20/(16+20) = 20/36 ≈ 0.556

Therefore:

Posterior mean = 0.444(0.3125) + 0.556(0.45)

Posterior mean ≈ 0.389

Mathematically, the prior receives approximately 44.4% of the weight, while the observed sample receives approximately 55.6% of the weight.

Why do the posterior mean and shrinkage matter?

The posterior mean gives us one understandable number with which to summarize the posterior distribution. Instead of describing every possible value of π, we can report 0.389 as our updated point estimate under this particular prior and model.

Shrinkage shows that Bayesian estimation does not simply replace the prior with the sample result. It balances the starting information with the new evidence. This can prevent a small or unusual sample from completely determining the estimate.

However, shrinkage also makes the choice of prior important. A strongly concentrated prior can pull the posterior estimate substantially toward the prior mean, especially when the sample is small. This is why we should justify our prior and compare results under multiple reasonable priors.

If we observed more movies while keeping the same prior, n would increase and the data would receive more weight. The posterior mean would then move closer to the observed sample proportion. If we used a more concentrated prior while keeping the sample size fixed, the prior would receive more weight and produce greater shrinkage toward its mean.

INTERACTIVE PLOT

Two bars will display the prior weight and the data weight. Readers will be able to change the sample size and prior concentration to see how the balance changes. A second display will show the prior mean, sample proportion, and posterior mean on the same scale, allowing readers to see the posterior estimate move between the prior belief and the observed result.

SECTION 13: FROM A POINT ESTIMATE TO A POSTERIOR DISTRIBUTION

The posterior mean provides one summary of the posterior, but Bayesian inference does not reduce the answer to a single number.

Under the lower-pass prior:

$\pi \mid y \sim \text{Beta}(14,22)$

The full Beta(14,22) distribution describes the remaining uncertainty about π after observing the movies.

An approximate 95% credible interval is:

(0.239, 0.551)

The Bayesian interpretation is:

Given the Beta(5,11) prior, the Binomial model, its assumptions, and the observed data, there is approximately a 95% posterior probability that π lies between 0.239 and 0.551.

This is different from the confidence-interval interpretation. A credible interval makes a probability statement about the unknown parameter, conditional on the model and prior.

The posterior can also answer questions about meaningful thresholds. For example:

$P(\pi < 0.50 \mid y) \approx 0.912$

Under this prior and model, there is approximately a 91.2% posterior probability that fewer than half of comparable movies pass.

This statement cannot be separated from its assumptions. It would be incomplete to report "there is a 91.2% probability" without identifying the prior, likelihood, population, and data on which that probability depends.

INTERACTIVE PLOT

Readers will select a threshold. The area of the posterior below or above that threshold will be shaded, and the corresponding posterior probability will be displayed.

SECTION 14: DOES THE PRIOR CHANGE THE CONCLUSION? *(Do we need this?)*

The three posterior means are:

Beta(5,11) prior → Beta(14,22) posterior → mean = 0.389

Beta(1,1) prior → Beta(10,12) posterior → mean = 0.455

Beta(14,1) prior → Beta(23,12) posterior → mean = 0.657

Their probabilities that π is below 0.50 are also different:

Under Beta(14,22):

$P(\pi < 0.50 \mid y) \approx 0.912$

Under Beta(10,12):

$P(\pi < 0.50 \mid y) \approx 0.668$

Under Beta(23,12):

$P(\pi < 0.50 \mid y) \approx 0.029$

The first analysis suggests that π is probably below 0.50.

The second suggests that π may be below 0.50, but the evidence is less decisive.

The third suggests that π is probably above 0.50.

All three analysts observed the same movies. Their conclusions differ because 20 observations are not enough to overwhelm sharply different prior beliefs.

This is why prior sensitivity analysis matters. A Bayesian analysis should not hide the effect of the prior. It should examine whether reasonable alternatives lead to substantially different conclusions.

If the conclusion remains stable across multiple defensible priors, it is relatively robust.

If the conclusion changes dramatically, the honest conclusion may be that the current data are not strong enough to settle the question.

SECTION 15: WHAT HAPPENS WHEN WE OBSERVE MORE MOVIES?

The influence of the data depends partly on the sample size.

Recall that the weight given to the data in the posterior mean is:

$n/(\alpha+\beta+n)$

As n increases:

$n/(\alpha+\beta+n)$ approaches 1

At the same time, the prior weight:

$(\alpha+\beta)/(\alpha+\beta+n)$

approaches 0.

Suppose two samples both have pass rates close to 45%.

A sample with 9 passes out of 20 provides some evidence about π, but its likelihood is fairly wide.

A sample with 90 passes out of 200 has the same observed proportion, but it provides much stronger evidence. Its likelihood is narrower because many more observations support the estimate.

As the sample grows, posteriors produced by different priors tend to move closer together. The data increasingly dominate the starting beliefs– the balance between the prior and data changes as the amount of evidence changes.

INTERACTIVE PLOT (?)

Hold the observed pass rate near 45% while increasing the sample size from 20 to 50, 100, and 500. The three posterior curves will gradually narrow and move toward one another.

SECTION 16: LEARNING ONE MOVIE AT A TIME *(Do I need this? Is sec 17 enough?)*

Bayesian learning can occur sequentially.

Suppose we begin with:

$\pi \sim \text{Beta}(5,11)$

The first observed movie passes. The updated distribution becomes:

$\pi \mid \text{first movie} \sim \text{Beta}(6,11)$

If the second movie fails:

$\pi \mid \text{first two movies} \sim \text{Beta}(6,12)$

Each passing movie adds one to the first parameter.

Each failing movie adds one to the second parameter.

After all 20 movies, including 9 passes and 11 failures, we obtain:

$\pi \mid \text{all 20 movies} \sim \text{Beta}(14,22)$

The posterior after one stage becomes the prior for the next stage.

Importantly, updating one movie at a time produces the same final posterior as updating with all 20 movies at once, assuming the same model and starting prior.

INTERACTIVE PLOT

Readers will reveal the sampled movies one at a time. After each pass or failure, the prior will transform into a new posterior. The graph will track the posterior mean and uncertainty throughout the sequence.

SECTION 17: PREDICTING FUTURE MOVIES

So far, we have used the posterior to learn about π. We can also use it to predict future movie outcomes.

Suppose Ynew is the number of passing movies in a future sample of m movies.

Conditional on π:

Ynew | π ~ Binomial(m, π)

But π is unknown. Instead of substituting one estimate and pretending it is exact, Bayesian prediction averages over the entire posterior distribution:

p(Ynew | y)

= ∫ p(Ynew | π)p(π | y)dπ

When the posterior is Beta, this produces a Beta-Binomial posterior predictive distribution.

Under the Beta(14,22) posterior, the expected number of passing movies in the next m movies is:

E(Ynew | y) = m × 14/36

For the next 10 movies:

E(Ynew | y) = 10 × 14/36 ≈ 3.89

This does not mean that exactly 3.89 movies will pass. The number is the posterior predictive mean. The complete predictive distribution assigns probabilities to 0, 1, 2, ..., 10 passing movies.

INTERACTIVE PLOT

Choose the number of future movies. The plot will display the predictive distribution for how many may pass. A simulation button will generate imaginary future samples from that distribution.

SECTION 18: BEYOND PASS OR FAIL—TWEETS AFTER A MOVIE RELEASE

Note: work in progress. Will describe more of Gamma-Poisson distribution

The Bechdel-test analysis involves binary outcomes: each movie either passes or fails. However, Bayesian inference is not limited to questions with two possible outcomes.

Suppose a movie studio wants to measure online attention immediately after a new movie is released. Let:

λ = the typical number of tweets mentioning the movie per minute during the first hour after its release.

Unlike the Bechdel outcome, the number of tweets is a count. In each one-minute interval, we might observe 0, 1, 2, or many tweets. A Poisson distribution is commonly used to model the number of events occurring during a fixed period:

$Y_i \mid \lambda \sim \text{Poisson}(\lambda)$

Here, Y_i is the number of tweets observed during minute i, and λ is the unknown average number of tweets per minute.

Before the movie is released, the studio may already have information from trailers, advance ticket sales, or similar movie releases. Its uncertainty about the positive rate λ can be represented using a Gamma prior:

$\lambda \sim \text{Gamma}(\alpha, \beta)$

In this parameterization, α is the shape parameter and β is the rate parameter. The prior mean is:

$E(\lambda) = \alpha/\beta$

The values of α and β could be selected using data from comparable movie releases.

The Gamma distribution is useful because λ must be positive. It is also conjugate to the Poisson likelihood. This means that combining a Gamma prior with Poisson data produces another Gamma distribution as the posterior.

Suppose the studio observes tweet counts from n one-minute intervals:

$Y_1, Y_2, \ldots, Y_n$

If the total number of observed tweets is:

ΣY_i

then the posterior is:

$\lambda \mid \text{data} \sim \text{Gamma}(\alpha + \Sigma Y_i, \beta + n)$

For example, suppose the numbers of tweets during the first five minutes are:

8, 11, 10, 13, and 8.

The total is:

$\Sigma Y_i = 50$

Therefore, a Gamma(α, β) prior would update to:

λ | data ~ Gamma(α + 50, β + 5)

The update resembles the Beta-Binomial update from the Bechdel example. The prior parameters represent the starting information, and the observed data are added to produce the posterior parameters. However, the two models answer different kinds of questions:

- The Beta-Binomial model estimates an unknown proportion from pass-or-fail data.
- The Gamma-Poisson model estimates an unknown rate from count data.

This example also demonstrates conjugacy more broadly. Conjugacy is not a special feature that applies only to Beta and Binomial distributions. Different types of data have different useful conjugate pairs:

Beta prior + Binomial data → Beta posterior

Gamma prior + Poisson data → Gamma posterior

The Poisson model assumes that tweets occur independently and at a constant average rate during the period being studied. In reality, online activity may spike after an announcement, a celebrity post, or a major scene becoming viral. Bots, retweets, and coordinated marketing may also violate the independence assumption. Restricting the analysis to a short, clearly defined period and specifying exactly what counts as a movie-related tweet would make the simplified model more defensible.

INTERACTIVE PLOT

Readers will adjust the Gamma prior and enter tweet counts from successive one-minute intervals. The graph will display the prior, likelihood, and posterior distributions for λ. A prediction panel could also show the posterior predictive distribution for the number of tweets expected during the next minute.

SECTION 19: COMPARING THE FREQUENTIST AND BAYESIAN RESULTS

The frequentist and Bayesian analyses begin with the same observation:

9 of the 20 sampled movies passed the Bechdel test.

However, the two approaches use this information differently and answer different types of questions.

Comparing the point estimates

The frequentist maximum likelihood estimate is:

$\hat{\pi}_{MLE} = y/n = 9/20 = 0.45$

This means that the value of π that makes the observed sample most likely is 0.45. The MLE is based only on the observed data; it does not incorporate prior beliefs or outside information.

The Bayesian analysis combines the data with a prior distribution. Therefore, its point estimate depends on the prior that was selected. Using the posterior mean as the Bayesian point estimate, we obtained:

- Lower-pass prior (i.e., moviegoer's prior belief was that most movies won't pass): posterior mean = 0.389
- Neutral prior (i.e., neutral belief): posterior mean = 0.455
- Higher-pass prior (i.e., prior belief that most movies will pass): posterior mean = 0.657

Under the lower-pass prior:

$\pi \sim \text{Beta}(5,11)$

The prior mean is:

$5/(5+11) = 0.3125$

After observing 9 passes and 11 failures, the posterior is:

$\pi \mid y \sim \text{Beta}(14,22)$

The posterior mean is:

$14/(14+22) \approx 0.389$

This estimate is below the sample proportion of 0.45 because the prior placed more weight on lower pass rates.

Under the neutral prior:

$\pi \sim \text{Beta}(1,1)$

The Beta(1,1) distribution assigns equal prior density to every possible value of π between 0 and 1. It does not favor low, middle, or high pass rates.

After observing the data, the posterior is:

$\pi \mid y \sim \text{Beta}(10,12)$

The posterior mean is:

$10/(10+12) \approx 0.455$

This is very close to the sample proportion because the Beta(1,1) prior is relatively weak compared with the information contributed by the 20 observed movies.

Under the higher-pass prior:

$\pi \sim \text{Beta}(14,1)$

This prior places most of its weight on high pass rates. After observing the data, the posterior is:

$\pi \mid y \sim \text{Beta}(23,12)$

The posterior mean is:

$23/(23+12) \approx 0.657$

Although only 45% of the sampled movies passed, the higher-pass prior pulls the posterior mean upward.

These results illustrate **prior sensitivity**. With only 20 observations, substantially different priors can produce noticeably different posterior estimates. If we observed a much larger sample, the data would receive more weight and the posterior estimates would generally move closer together.

Comparing the intervals

A point estimate provides one possible summary, but it does not show the range of values that remain plausible. Both frameworks use intervals to communicate uncertainty, although the intervals have different interpretations.

The 95% Wilson confidence interval from the frequentist analysis is approximately:

(0.258, 0.658)

As a practical summary, values of π between approximately 0.258 and 0.658 are reasonably compatible with the observed data under the frequentist model.

The formal interpretation concerns the interval-building procedure. If we repeatedly selected samples in the same manner and constructed a 95% Wilson interval from each sample, approximately 95% of those intervals would contain the fixed, unknown value of π.

This does not formally mean that there is a 95% probability that π lies between 0.258 and 0.658. In frequentist inference, π is fixed. The interval changes from one hypothetical sample to another.

Under the lower-pass Beta(5,11) prior, the 95% Bayesian credible interval is approximately:

(0.239, 0.551)

The Bayesian interpretation is:

Given the Beta(5,11) prior, the Binomial model, and the observed data, there is a 95% posterior probability that π lies between approximately 0.239 and 0.551.

The confidence and credible intervals overlap, but they do not mean the same thing:

- The confidence interval describes the long-run performance of a procedure applied to repeated samples.
- The credible interval describes posterior uncertainty about π after conditioning on the selected prior, model, and observed data.

The Bayesian interval is shifted toward lower values because it was calculated using the lower-pass prior.

Comparing the hypothesis test and posterior probability

The frequentist analysis tested:

H_0: $\pi = 0.50$

H_a: $\pi < 0.50$

The one-sided p-value was:

$P(Y \leq 9 \mid \pi = 0.50) \approx 0.412$

This means that if the true pass rate were exactly 0.50, there would be approximately a 41.2% probability of observing 9 or fewer passes in a sample of 20.

Because 41.2% is not a small probability, the observed result is not particularly unusual under the null hypothesis. Therefore, the sample does not provide strong frequentist evidence that π is below 0.50.

The p-value is not the probability that the null hypothesis is true. In particular:

$\text{p-value} \neq P(\pi = 0.50 \mid \text{data})$

The Bayesian analysis asks a different question. Under the Beta(5,11) prior and the resulting Beta(14,22) posterior:

$P(\pi < 0.50 \mid y) \approx 0.912$

Given this prior, model, and sample, there is approximately a 91.2% posterior probability that fewer than half of comparable movies pass the Bechdel test.

At first, the p-value of 0.412 and posterior probability of 0.912 may appear contradictory. However, they are probabilities of different events:

- The p-value asks: If π were exactly 0.50, how likely would it be to observe 9 or fewer passes?
- The posterior probability asks: After observing the data, and given the selected prior and model, how probable is it that π is below 0.50?

The frequentist calculation conditions on the null hypothesis and evaluates the data. The Bayesian calculation conditions on the observed data and evaluates possible values of π.

Therefore, 0.412 and 0.912 are not competing answers to the same question. In addition, the Bayesian result depends on the prior. A different prior would produce a different value for $P(\pi < 0.50 \mid y)$.

SECTION 20: WHAT COULD GO WRONG?

Real-world interpretation requires care.

First, the sample contains only 20 movies. With such a small sample, the likelihood is not strong enough to eliminate disagreement among substantially different priors.

Second, the three priors are teaching tools. They represent hypothetical viewpoints from the textbook, not prior distributions estimated from historical movie studies.

Third, the source dataset does not represent every movie ever made. It may overrepresent certain years, genres, countries, budgets, or commercially prominent movies.

Fourth, one common π may be too simple. The probability of passing could change across time or differ by genre, studio, country, budget, or intended audience.

Fifth, the Binomial model assumes independent observations with a common pass probability. Movies produced by the same studio, creative team, franchise, or historical period may be related.

Sixth, Bechdel classifications may involve judgment. Viewers might disagree about whether a character is named, whether two women genuinely speak to one another, or whether a conversation is about something other than a man.

Seventh, the analysis describes representation according to one narrow test. It does not measure the quality, complexity, importance, or screen time of female characters.

Finally, this analysis does not establish causation. It cannot show that passing the Bechdel test causes a movie to earn more money, receive better reviews, or influence audiences differently.

A more advanced project could model pass probability as a function of year or genre. That would require Bayesian regression or hierarchical modeling and is beyond the main introductory analysis.

SECTION 21: WHAT FREQUENTIST AND BAYESIAN INFERENCE ADD

We began with one observed result:

9 of 20 sampled movies passed the Bechdel test.

The sample proportion was:

9/20 = 0.45

This number alone does not describe our uncertainty, tell us which other values of π remain plausible, or explain how prior information might affect our conclusion.

The frequentist analysis treated π as a fixed but unknown parameter. It used the sample to:

- estimate π using the MLE,
- identify compatible values using a confidence interval,
- and evaluate whether the data were unusual under a specific null hypothesis.

The Bayesian analysis represented uncertainty about π using a probability distribution. It combined:

Prior × Likelihood → Posterior

For the Bechdel example:

Beta prior × Binomial likelihood → Beta posterior

This allowed us to calculate posterior estimates, credible intervals, and direct probabilities about π. It also made the influence of different starting beliefs visible.

The three moviegoers began with different priors, so they did not reach identical posterior conclusions. Nevertheless, observing the same 9 passes and 11 failures moved all three toward the evidence in the sample. This demonstrated a central feature of Bayesian learning: both the amount of data and the strength of the prior determine the posterior.

We also considered the number of tweets posted after a movie's release. This showed that Bayesian reasoning is not limited to binary pass-or-fail outcomes:

- The Beta-Binomial model estimated an unknown proportion.
- The Gamma-Poisson model estimated an unknown event rate.

Although the distributions and parameters changed, the underlying Bayesian process remained the same: select a model, represent prior uncertainty, observe data, and update the prior to obtain a posterior.

Frequentist and Bayesian inference therefore provide different but useful descriptions of uncertainty. Frequentist inference evaluates estimates and procedures through their behavior over repeated samples. Bayesian inference updates a probability distribution for an unknown parameter, conditional on the prior, model, and observed data.

Neither approach makes assumptions disappear. A responsible conclusion must still explain:

- what population the parameter represents,
- how the data were collected,
- which assumptions were made,
- how much uncertainty remains,
- and how sensitive the conclusion is to the chosen method or prior.

The central lesson is not simply that the estimated pass rate is 0.45. It is that statistical conclusions depend on the evidence, the model, and the framework used to interpret uncertainty.